

Question#1 [Coding] [12 Marks]

You are given an $N \times N$ 2D array. You must store only the **positive** numbers in a **dummy-headed singly linked list**, where each node contains (**value**, **row_index**, **column_index**).

Traverse the matrix in **row-major order** (row by row from top to bottom, and within each row from left to right). For every **positive** element, create a node and insert it into the list using the rule below:

- In the linked list, nodes that satisfy the condition **value + (row_index + col_index) > threshold** must be placed **before** the nodes that do not satisfy the condition.
- The **relative order of appearance in the matrix must be preserved** within each of the two groups (those that satisfy the condition and those that do not).

Write a function that takes a 2D matrix **M**, an integer **threshold**, and a dummy head **dh** as parameters. The function should build the singly linked list according to the rules given and return the dummy head **dh**.

Note: Do not create additional arrays or linked lists. Also, assume that the matrix size will always be at least 2×2 .

Java Custom Node Class (Given)		Python Custom Node Class (Given)
<pre>class Node { int value; int row_index; int column_index; Node next; Node(int value, int row_index, int column_index){ this.value = value; this.row_index = row_index; this.column_index = column_index; this.next = null; } }</pre>		<pre>class Node: def __init__(self, value, row_index, column_index): self.value = value self.row_index = row_index self.column_index = column_index self.next = None</pre>
Java Method Signature		Python Method Signature
<pre>static Node buildAndRearrange(int[][] M, int threshold, Node dh){ // to do }</pre>		<pre>def build_and_rearrange(M, threshold, dh): #to do</pre>
Sample Given Parameters		Sample output
Given Matrix	Given Threshold	dh -> (5,0,2) -> (6,2,0) -> (4,0,0) -> (3,1,1) -> (1,2,2) -> null
4 -1 5 0 3 -2 6 -3 1	6	
Explanation: Index (0,0) = 4 (positive) → Node [4,0,0]: 4 + (0+0) = 4 > 6 → False Index (0,1) = -1 (negative) → Skip Index (0,2) = 5 (positive) → Node (5,0,2): 5 + (0+2) = 7 > 6 → True Index (1,0) = 0 (neutral) → Skip Index (1,1) = 3 (positive) → Node (3,1,1): 3 + (1+1) = 5 > 6 → False Index (1,2) = -2 (negative) → Skip Index (2,0) = 6 (positive) → Node (6,2,0): 6 + (2+0) = 8 > 6 → True Index (2,1) = -3 (negative) → Skip Index (2,2) = 1 (positive) → Node (1,2,2): 1 + (2+2) = 5 > 6 → False Final linked list: dh -> (5,0,2) -> (6,2,0) -> (4,0,0) -> (3,1,1) -> (1,2,2) -> null) Because the nodes (5,0,2) and (6,2,0) satisfy the condition, they are placed before the nodes (4,0,0), (3,1,1), (1,2,2), which do not satisfy the condition.		

i) What is the time complexity to delete a node at the end of a Singly Linked List? [1 mark] Answer:		
ii) True or False: A Doubly Linked List requires more memory per node than a Singly Linked List. [1 mark] Answer:		
iii) Assume you're given a singly linked list: dh -> 1 -> 2 -> 3 -> 4 -> 5 -> null Draw the resulting linked list after executing the code below: [1 mark]		iii) Answer:
Java:	Node current = dh.next; current.next = current.next.next.next;	
Python:	current = dh.next current.next = current.next.next.next	
iv) Consider the following statements about 2D arrays. Identify the FALSE statement. [1 mark] (A) The total number of elements in an N×M shaped array is N multiplied with M. (B) Iterating over a 2D array typically requires nested loops. (C) Time Complexity of arr[0][0] and arr[99][99] is different. (D) You can represent images or tables using 2D arrays.		v) If arr is a 5×6 2D array, which expression gives the number of columns in the 3rd row? [1 mark] (A) arr.length (B) arr[2].length (C) arr[4].length (D) arr.length * arr[2].length
vi) Given the matrix: <pre>[[1 2] [3 4] [5 6]]</pre>	What is the output of the following code: [1 mark] <pre>for (int i = 2; i >= 0; i--) { for (int j = 1; j >= 0; j--) { print(matrix[i][j]); } }</pre>	Answer:

Question#2 [Short Questions] [6 marks]

You can start answering Question#1 from here: